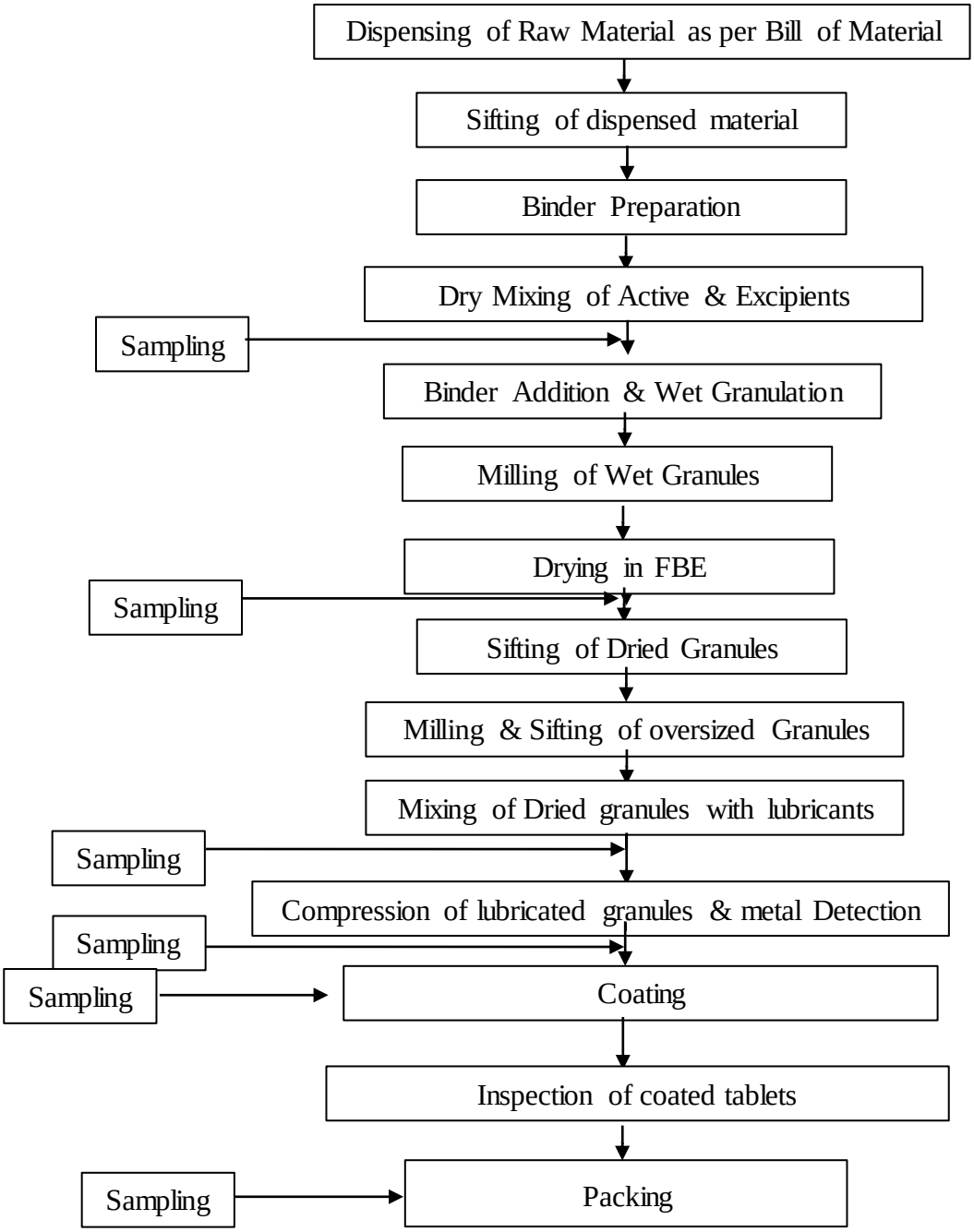
P 3.3 Manufacturing Process and Process Control



A. DESCRIPTION OF EQUIPMENT, FACILITY AND UTILITIES INVOLVED IN MANUFACTURING PROCESS

All the major equipments and instrument used for process, as listed below shall be verified for Installation, Operational and Performance Qualification.

Sr. No.	Name of the equipment	Equipment ID
Warehouse		
01	OFCB	EQ/WRH/OFCB/004
02	Weighing Balance	INQ/WRH/WEB/003
03	Weighing Balance	INQ/WRH/WEB/004
Production		
01	Double Deck Vibratory Sifter (30 inch)	EQ/OSD/VSF/002
02	High Shear Mixer Granulator - 100L	EQ/OSD/HSMG/001
03	Pillar Type Bin Blender - 300 L	EQ/OSD/PBB/001
04	Binder Preparation Vessel 60 L	EQ/OSD/BPV/001
05	Fluid Bed Equipment (FBE 125 C)	EQ/OSD/FBE/001
06	Tap Cone Mill -100	EQ/OSD/TCM/001
07	Weighing Balance	INQ/OSD/WEB/014
08	Weighing Balance	INQ/OSD/WEB/013
09	27 Station 'D' Tooling Double Rotary Compression Machine	EQ/OSD/COM/001
10	Combo metal detector and de-duster	EQ/OSD/DMD/001 EQ/OSD/DMD/002
11	Weighing Balance	INQ/OSD/WEB/003
12	Capsule and tablet inspection belt	EQ/OSD/CTIC/001
13	Blister Packing Machine	EQ/OSD/RBPM/001
14	Weighing Balance	INQ/OSD/WEB/005
IPQA		
01	Weighing Balance	INQ/IPQA/WEB/001
02	Vernier Caliper	INQ/IPQA/VC/001
03	Disntegration Test Apparatus	INQ/IPQA/DT/001
04	Leak Test Apparatus	INQ/IPQA/LT/001
05	Hardness Tester	INQ/IPQA/THT/001
06	Halogen Moisture Analyzer	INQ/IPQA/HMA/001
07	Friability Test apparatus	INQ/IPQA/FT/001

Sr. No.	Name of the equipment	Equipment ID
Quality Control		
01	Analytical Weighing Balance	INQ/QCD/WEB/002, INQ/QCD/WEB/003
02	HPLC	INQ/QCD/HPLC/001,INQ/QCD/HPLC/002 INQ/QCD/HPLC/006.
03	UV Visible Spectrophotometer	INQ/QCD/UV/001
04	IR	INQ/QCD/IR/001
05	Vernier Caliper	INQ/QCD/VC/002
06	Disintegration test Apparatus	INQ/QCD/DT/001
07	Friability Test apparatus	INQ/QCD/FT/001
08	Hardness Tester	INQ/QCD/THT/001
09	pH meter	INQ/QCD/PH/001
10	Tap Density Apparatus	INQ/QCD/TAP/001
11	Dissolution test apparats	INQ/QCD/DIS/001, INQ/QCD/DIS/002
12	Autotitrator	INQ/QCD/AT/001
13	UV Cabinet	INQ/QCD/UVC/001
14	Stability Chember	EQ/QCD/STB/001 to EQ/QCD/STB/003
Microbiology		
01	Weighing Balance	INQ/QCM/WEB/001, INQ/QCM/WEB/002
02	HLAF	EQ/QCM/HLAF/001
03	Vertical Autoclave	EQ/VA/001
04	Biosafety Cabinet	EQ/QCM/BSC/001

B. BRIEF DISCRIPTION OF MANUFACTURING PROCEDURE

In response to a requisition from Production, the raw materials warehouse personnel assemble the requisite raw materials in whole containers, on a batch specific pallet in the raw material staging area. Materials are verified as to correct stock rotation, presence of "Passed" labels from quality control, identification, and quantity.

The pallet is transferred into the materials dispensary. Prior to transfer, the dispensary area is checked for cleanliness, and absence of any other raw materials from a previous dispensing operation. Upon receipt in the dispensary area, production staff and Quality Assurance staff verifies quality control status, batch number, identity, and quantities of materials received, against the accompanying documents.

Precaution: -

1. Ensure visually the cleanliness of production area, machines & equipment's.
2. Ensure visually that previous product is completely removed.
3. Ensure that the residual test clearance report is obtained from QC.
4. Check temperature & humidity of area.
5. Every person carrying out the manufacturing process in production shall wear the protective clothing, cap, gloves, and nose mask whenever required.
6. Ensure all material received is appropriately labeled. Verified for Net weight & Appearance by Competent Technical Person.
7. Ensure the material taken for processing are free of Black particles / foreign matters.
8. Cover the containers properly at all time. Upon receipt in the production area, a further independent verification of material identity, quantity, batch number and release status is performed by production staff.

Stage	Area	Temperature	Relative Humidity	Pressure Differential
Dispensin	Dispensing Area	22°C±3°C	50% ± 10	5-25 Pascal
Filling	Capsule Filling Room	22°C±3°C	50% ± 10	5-25 Pascal

1. Sifting:	
1.1	Solifenacin Succinate & lactose monohydrate geometrically mixed & then mixed in polybag. 1:1 proportion 3 times.
1.2	Add mixture of solifenacin succinate and Lactose monohydrate and maize starch trough sieve # 40

1.3	Sift Magnesium stearate through 60# sieve & keep aside for lubrication stage.
2. Dry Mixing:	
2.1.	Load the sifted materials (Solifenacin Succinate+lactose monohydrate+maize starch starch+Magnesium stearate) into the RMG. Dry mix the material for 10 minutes at impeller slow speed.
3. Binder Preparation	
3.1	Check the cleanliness of the solution preparation vessel.
3.2	Take 1.6656 Kg purified water in binder preparation vessel then add slowly hypromellose HPMC 3mpas with continuous stirring to form a clear solution.
4. Granulation:	
4.1	Add the binder solution slowly into RMG at impeller slow speed mix for 1 minutes followed by racking.
4.2	If required add 0.7992 kg of purified water in RMG at impeller slow speed mix for 1 minute follwed by racking.
4.3	Start the kneading by operating the impeller at Fast Speed for 30 seconds.
4.4	Unload the material.
5. Drying:	
5.1	Check the integrity of the dryer bowl before and after drying
5.2	Dry the granules in dryer at inlet temperature 50°C to 60 °C. Drying to be done till desire LOD achieved (LOD Limit 1.0 ±2.0 % at 105 deg. Cel.).
5.3	Record the observations
6. Sizing & milling	
6.1	Dried granules size through 20# sieve and oversized granules milled through 3, 2.5, and 2 mm screen simultaneously by the Tap cone mill, 20# retention from each milling stage should be milled by subsequent screen. At last, retaintion after 2 mm screen milling observed should be pressed out through 20# sieve. Final granules size: 20#
6.2	Calculate the yield of dried granules.
7. Lubrication:	
7.1	Load the 20# sized above granules & pre sifted magnesium stearate in blender and lubricate for 5 minutes at 08 RPM.
8. Compression	
8.1	Temperature NMT 22±3°C and Humidity NMT 50% ±10% of the area shall be maintained

8.2

Punch details	7.6 mm Round, standard concave punches plain on both sides ('B' Toolings)
Appearance	White to off white, round shaped, film coated tablet plain on other sides.
Group weight of 20 tablets	16.400 gm ± 3.0% (15.908 gm to 16.892 gm)
Average Weight	150.00 mg ± 7.5% (138.75 mg to 161.25 mg)
Thickness	3.3 mm ± 0.30 mm (3.00 mm to 3.6 mm)
Hardness	NLT 3.0 kg
Friability (%)	NMT 1.00 % w/w
Disintegration Time	Not more than 15 minutes
Speed	16-42 RPM

8.3	Collect tablets completion of one rotation each from left & right side discharge chute of machine for initial checks.
8.4	Run the machine after initial checks of tablet parameters are found satisfactory.
8.5	Record the weight of compressed tablets.
8.6	After inspection, carry out the challenge test before starting the metal detection.
9. Coating solution preparation	
9.1	Take the 3.510 kg purified water add slowly Ready coat 31FO193220 under stirring in solution preparation vessel under vertex forming stirring & stir for 45 minutes to form uniform coating dispersion. Filter the coating solution before use by 100# sieve.
9.2	After getting the solution, filter the solution through SS screen.
10. Film coating:	
10.1	Preheat the tablets for 5 min. after reaching inlet air temperature 70 ºC ± 10ºC
10.2	Set the nozzle spray tip between 8 ± 4 inches for Auto-coater from moving tablet bed.
10.3	Bed Temperature should be 38°C ± 7°C

10.4 Standard Setting Parameters of coating

Parameter	**Acceptance criteria/Limits**
Inlet Temperature (°C)	60°C to 80 °C
Exhaust Temperature (°C)	45°C to 60 °C
Bed Temperature (°C)	38 ± 10 °C
Pan speed	2-8 RPM
Automizing Air pressure kg/cm^2	0.1 – 0.4
Peristaltic pump speed (RPM)	5 – 40

<table>
<tr><td></td><td>Note: Coating parameters can be adjusted to achieve better coating.</td></tr>
<tr><td>10.5</td><td>Once target weight gain of 2.0% is achieved stop the spraying process and dry the tablet for approximately 5 minutes.</td></tr>
<tr><td>10.6</td><td><table>
<tr><th>Parameter</th><th>Acceptance criteria/Limits</th></tr>
<tr><td>Appearance</td><td>Light Yellow coloured, round shaped, film coated tablet plain on other sides.</td></tr>
<tr><td>Weight of 20 coated tablets</td><td>16.720gm ± 3.00 % (16.218gm – 17.222gm)</td></tr>
<tr><td>Average weight of coated tablet</td><td>153.00 mg ± 7.5% (141.53 mg to 164.48 mg)</td></tr>
<tr><td>Thickness</td><td>3.28 mm ± 0.3 mm (2.98 mm to 3.58 mm)</td></tr>
<tr><td>Disintegration Time</td><td>NMT 30min.</td></tr>
</table>
After coating check the following tablets parameters.</td></tr>
<tr><td colspan="2">11. Inspection coated tablets:</td></tr>
<tr><td>11.1</td><td>Inspect the coated tablets for any defect.</td></tr>
<tr><td>11.2</td><td>Record the weight of inspected tablets after completion of inspection.</td></tr>
<tr><td>11.3</td><td>Calculate % yield. (Limit is 95.00% to 100.00%)</td></tr>
</table>